Heriot Watt University,
Department of Mathematics

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Assessed Coursework 1

Module F18CD 2025–2026

Handed out: Friday 3rd October 2025 (2PM UK time)

Due: Friday 10th October by:

- **Edinburgh: 2PM**
- **Malaysia: 9PM**
- **Dubai: 5PM**
- You must submit your work via the Gradescope submission link available on assignment page in Canvas.
- Start each question on a new page (Parts of the same question can be on the same page).
- This assignment contains 2 questions, totalling 20 marks.
- Unless stated otherwise, you may utilise any theorems proven in the lecture notes. Make sure to state explicitly which theorems you use from the notes in your answers.

Questions

1. Let A be a nonempty subset of $\mathbb{R}$ that is bounded above and let $c \in \mathbb{R}$. Let

$$cA := \{ca : a \in A\}.$$

(a) Prove that $\sup(cA) = c \sup A$ if $c \geq 0$.

(b) Guess the value of

$$\sup\left\{\frac{2n-5}{3n} : n \in \mathbb{N}\right\}$$

and prove that your guess is correct. Hence, or otherwise, state

$$\sup\left\{\frac{4n-10}{n} : n \in \mathbb{N}\right\}.$$

(c) State whether the result in (a) is true or false if $c < 0$. Provide a proof if it is true, or otherwise give a counterexample.

[9 marks]

2. Let (x_n) and (y_n) be sequences such that

$$y_n = \frac{x_1 + x_2 + \cdots + x_n}{n}, \quad \text{for all } n \in \mathbb{N}.$$

(a) Prove that if (x_n) is convergent, then (y_n) is convergent to the same limit.

(b) Show that the sequence (z_n) given by

$$z_n = \frac{1}{n} \sum_{k=1}^{n} \frac{1}{k}$$

is convergent.

(c) Determine whether the converse of (a) is true or false. Provide a proof if it is true, or otherwise give a counterexample.

[11 marks]